

Low-power sensor networks for grid connection compliance monitoring of distributed generators.

Energy × Regulatory compliance and reporting × LPWAN (LoRaWAN, NB-IoT, LTE-M) · OS056 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
66 is the world moving	46 can we play, can we win	MEDIUM 0 named in the evidence	€171m partial evidence · per year	NOW budgeted procurement within 1...	L1 13 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

Evidence gap (SC-13). Orange has 3.0 published reference(s) in Energy, below the threshold of 5. A customer conversation here has no proof point behind it yet.

The opportunity

This opportunity addresses the need for low-power sensor networks to monitor grid connection compliance for distributed generators, driven by the EU's revision of the Network Code on Requirements for Grid Connection of Generators. It targets energy producers and grid operators who must adapt to new regulatory requirements. The opportunity exists now because the regulatory change is underway, creating an immediate compliance imperative.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€207m €32.7m – €3.2bn	€171m €27.0m – €2.6bn	€2.1m €324k – €31.7m	partial
Observed: tendered contract value, annualised	€172m €172m – €460m	€172m €172m – €460m	€2.1m €2.1m – €5.5m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Energy (EU27_2020)	5,796 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting LPWAN (LoRaWAN, NB-IoT, LTE-M)	51.7 % of enterprises (10+ employees)	proxy	Eurostat · isoc_eb_iotn2 · E_IOT1	2021
Annual value of one engagement	€299k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Size-weighted buyer base	1,341 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Internet of Things by NACE Rev. 2 activity series E_IOT1 is used as a proxy for LPWAN (LoRaWAN, NB-IoT, LTE-M): IoT adoption as the connectivity demand proxy. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

Observed: tendered contract value, annualised

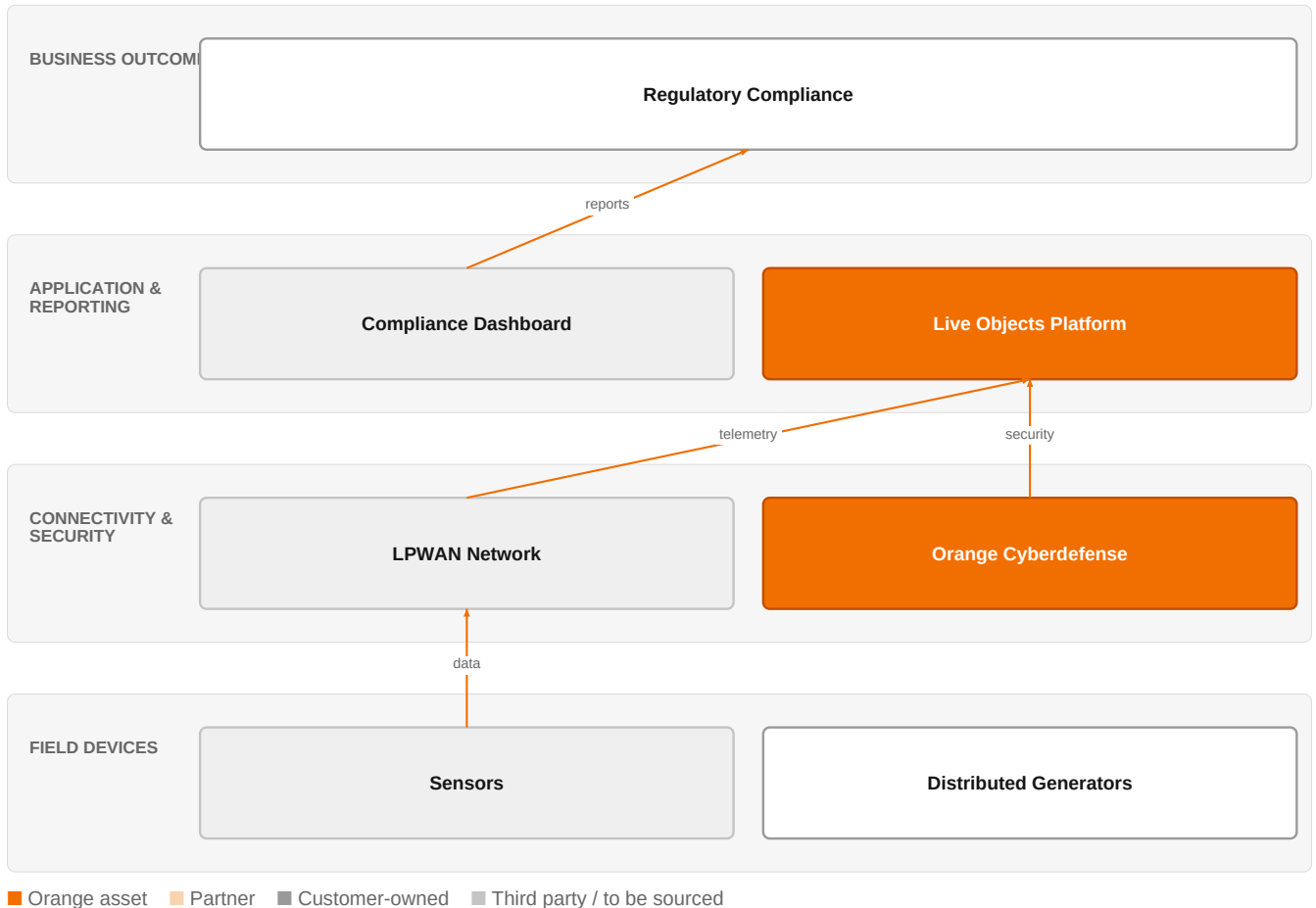
Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€172m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-19..2026-08-07
Assumed obtainable share of the serviceable market	1.2 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 21 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a

market size.

The solution, in one picture

Grid Compliance Monitoring Solution



This solution provides a secure, end-to-end monitoring chain from sensors to compliance reporting, leveraging Orange's IoT platform and managed security.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The EU is revising the Network Code on Requirements for Grid Connection of Generators, which will impose new compliance obligations on energy producers and grid operators. This regulatory shift is the primary catalyst, as it mandates more rigorous monitoring and reporting of grid connection parameters. Additionally, the market is seeing increased adoption of LPWAN technologies (LoRaWAN, NB-IoT, LTE-M) for such monitoring, as evidenced by recent tenders for smart metering and sensor systems in utilities and infrastructure. The technology is mature enough to support low-power, wide-area connectivity for distributed assets.

Sources: SIG-A8CE3660ACBA ec.europa.eu 2026-07-08; SIG-B13108F11065 ted.europa.eu 2026-07-07; SIG-B076E0F3616E ted.europa.eu 2026-05-20

Who buys, and why now

The primary buyer is the head of grid operations or regulatory compliance officer at a distribution system operator (DSO) or a large energy producer with distributed generation assets. The pain is operational: they need to demonstrate compliance with the revised grid connection code, which requires continuous monitoring of voltage, frequency, and other parameters at the point of connection. The trigger is the publication of the revised network code, which sets a deadline for compliance. They are also motivated by the risk of penalties for non-compliance and the need to avoid grid instability.

Why it is hot now — every claim bound to a dated source

- The EU is revising the Network Code on Requirements for Grid Connection of Generators, affecting energy producers and grid operators. [SIG-A8CE3660ACBA]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution using its IoT experts and Digital experts to design and deploy a sensor network based on LPWAN technologies. The engagement would start with a compliance assessment to identify monitoring points, then deploy sensors connected via Live Objects, Orange's IoT platform, to collect and manage data. Orange Cyberdefense managed services would be layered on to secure the network and data, ensuring compliance with ISO 27001 and SecNumCloud certifications. The solution would include ongoing monitoring and reporting to meet regulatory requirements.

Why Orange can win

Orange can win because it combines IoT expertise with strong security credentials, including ISO 27001 and SecNumCloud certifications, and offers a managed service via Orange Cyberdefense. The ability to provide a secure, end-to-end solution from sensors to reporting is a differentiator. However, the assets are not unique; competitors also offer LPWAN connectivity and security. Orange's advantage lies in its integrated approach and sovereign, certified infrastructure, which is particularly relevant for energy infrastructure.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Live Objects	offer	L1	offer provides the technology in a matching domain	unconfirmed
Orange Cyberdefense managed services	offer	L1	offer addresses the use case but not this technology	unconfirmed
ISO 27001	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
SecNumCloud	certification	SUP	certification Orange holds is required by this vertical	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 4.8; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include telecom operators like Telia Company, Vodafone Business, and Telefonica Tech, who offer LPWAN connectivity and compete on network coverage and pricing. Industrial/OT specialists like Actility, Honeywell, Schneider Electric, and Siemens have deep domain expertise in energy and can provide integrated solutions. Global systems integrator Capgemini is also active in energy. The customer will compare based on total cost, reliability, security, and the ability to meet regulatory requirements. Orange must differentiate on security and managed services.

Sources: SIG-A8CE3660ACBA ec.europa.eu 2026-07-08

Competitor	Category	Position here	Basis	Evidence
Telia Company	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Vodafone Business	Telecom operator peer	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Actility	Industrial / OT specialist	sells LPWAN (LoRaWAN, NB-IoT, LTE-M); competes in Connectivity Solutions, OX: Smart Industries	structural	—
Capgemini	Global systems integrator	active in Energy; competes in OX: Smart Industries	structural	—

Telefonica Tech	Telecom operator peer	active in Energy; competes in Connectivity Solutions	structural	—
Honeywell	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Schneider Electric	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—
Siemens	Industrial / OT specialist	active in Energy; competes in OX: Smart Industries	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 What is your current process for monitoring grid connection compliance for distributed generators?
- 2 Have you assessed the impact of the revised EU Network Code on your operations?
- 3 What is the scale of your distributed generation assets and their geographic spread?
- 4 Do you currently use any LPWAN-based monitoring solutions? If so, which ones?
- 5 What are your key concerns regarding data security and sovereignty for grid monitoring data?
- 6 What is your timeline for achieving compliance with the new regulations?

Objections you will meet

They will say	You can answer
We already have monitoring systems in place; why do we need a new one?	That's true, but the revised network code may require more granular and continuous data than your current systems provide. Our solution can complement or upgrade your existing infrastructure, and we can integrate with what you have.
LPWAN is not reliable enough for critical grid monitoring.	LPWAN technologies have matured significantly and are used in many industrial applications. We can design a network with redundancy and use our Orange Cyberdefense managed services to ensure reliability and security.
We are concerned about the cost of deploying sensors across many sites.	The cost is offset by the low power consumption and long battery life of LPWAN sensors, which reduce maintenance and operational costs. We can also phase the deployment to spread the investment.
Why should we choose Orange over a specialist like Siemens or Schneider?	Orange brings a unique combination of IoT expertise, secure connectivity, and managed security services, backed by sovereign certifications. We can provide an end-to-end solution that is both compliant and secure, and we can leverage our experience in large-scale deployments.

Risks and unknowns

The main risk is regulatory delay or changes in the network code requirements, which could stall investment. There is also uncertainty about the customer's existing infrastructure and whether they already have monitoring systems in place. The technology is proven, but integration with legacy systems may be complex. Additionally, the competitive landscape is crowded, and price pressure could erode margins. What is not yet known is the specific technical requirements of the revised code and the timeline for enforcement.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European grid operator, outlining how Live Objects and Orange Cyberdefense managed services can deliver a secure, compliant low-power sensor network, supported by ISO 27001 and SecNumCloud certifications.
Sales	In a conversation with a European grid operator's head of network operations, mention Orange's Live Objects platform and ISO 27001 certification as a foundation for secure, compliant monitoring solutions.
Strategist / Innovator	Commission a deep-dive brief on LPWAN-based grid compliance monitoring, leveraging Orange Group researchers and IoT experts to map the regulatory landscape and identify the most promising technical approach.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-A8CE3660ACBA	2026-07-08	ec.europa.eu	1	Revision of the Network Code on Requirements for Grid Connection of Generators
SIG-B13108F11065	2026-07-07	ted.europa.eu	1	Czechia – Energy meters – „Instalace Smart Meteringu na vodovodni siti – SVaK Ivancice“
SIG-D8CAA774FDC4	2026-07-01	Next Research	1	Securing 5G network slicing for the smart grid network
SIG-3164F57D73A7	2026-06-10	ted.europa.eu	1	Poland – Electricity distribution and control apparatus – Dostawy zabezpieczen pola dla Stacji ...
SIG-67F7CBD007F7	2026-05-22	Applied Sciences	1	Deployment and Coverage Optimization Methods for Base Stations Under Multi-Type Terminal Scenar...
SIG-B076E0F3616E	2026-05-20	ted.europa.eu	1	Italy – Sensors – DAC.0129.2025-Fornitura e posa di sistemi di monitoraggio per ponti e viadott...
SIG-C201BF32AE96	2026-01-07	openalex.org	1	Federated Learning for Secure Industrial Automation and Grid Optimization
SIG-1BE0A6CA2278	2026-01-01	openalex.org	1	Reinforcement Learning-Guided Federated Multi-Task Learning for Adaptive Energy Management in I...
SIG-1C6BA6D836AF	2025-12-17	openalex.org	1	FFT-Guided Multi-Window USAD with DTW–Isolation Forest for Reliable Anomaly Detection in Indust...
SIG-395E0DE089C6	2025-11-12	openalex.org	1	Adaptive Federated Learning for Industrial Wireless Energy Optimization
SIG-5EE745F73CED	2025-10-08	openalex.org	1	Multiview Spatial–Temporal Interaction Attention- Based Multivariate Time Series Anomaly Detect...
SIG-0D07056A550F	2025-10-01	cordis.europa.eu	1	Persistent Awareness system facilitating Customs CheckS
SIG-2424954F4021	2026-05-24	arxiv.org	3	ASTRO: Adaptive Spatio-Temporal Reinforcement Optimization for GNN Powered Anomly Detection in ...

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

Removed by those checks in this brief: diagram (1 box(es) claimed to be an Orange asset that was not supplied — shown as third-party instead)

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:15+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-18T20:43:46+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.